MATH 1210 Worksheet 4 Summer 2023

Suggested exercises, more exercises can be found in the textbook or on Dr. R.S.D. Thomas' webpage for archived material:

`http://home.cc.umanitoba.ca/~thomas/Courses`

1. Write in simplified Cartesian form
$$\frac{(1-i)^{14}}{(\sqrt{3}+i)^{10}}.$$

2. Find all solutions to $z^5 = 16(1-i)^2$. Write the solutions in exponential form and use principal values of the argument.

3. Consider the polynomial $P(x) = x^3 + 10x^2 - 37x + 26$.
   (a) Use Descartes rule of sign to determine the number of possible positive and negative real roots.
   (b) It is given to you that -13 is root. Find all other possible rational roots. Reduce the list as much as possible using (a).
   (c) Write $P(x)$ as a product of linear factors.

4. Consider the polynomial
$$P(x) = -x^3 + \frac{2}{3}x^2 - 2x + \frac{3}{2}.$$
   (a) Use Bounds theorem to find an upper bound on the absolute values of the roots.
   (b) Show that there are no roots in the interval $[10, 100]$.
   (c) Find a list of rational roots by using the rational root theorem. Use parts (a) and (b) to reduce the list. *Hint: clear the denominators.*

5. Consider the matrices
$$A = \begin{pmatrix} 1 & -1 \\ 2 & 3 \end{pmatrix} \qquad B = \begin{pmatrix} -2 & 4 \\ 1 & 0 \end{pmatrix}.$$
   Verify that AB and BA are not the same, by computing the two products.

6. Consider the matrices
$$A = \begin{pmatrix} 3 & 2 & 0 \\ -1 & 1 & -2 \\ 1 & 2 & -3 \end{pmatrix} \qquad B = \begin{pmatrix} 1 & 3 & 0 \\ 2 & 5 & -2 \end{pmatrix} \qquad C = \begin{pmatrix} 4 & -5 \\ 7 & 0 \\ 3 & -2 \\ 1 & -1 \end{pmatrix}$$
$$D = \begin{pmatrix} -1 & 2 & 1 & -2 \\ 0 & -1 & 1 & 0 \end{pmatrix} \qquad E = \begin{pmatrix} 1 & 3 \\ 2 & -2 \end{pmatrix}.$$
   Evaluate the following expressions if they exist or explain why they are undefined.
   (a) AB^TE
   (b) BA^TE
   (c) $A + C$
   (d) $2DB^T - 3E$
   (e) $2D - C^T$

7. Find all the values t such that the matrix
$$A = t \cdot \begin{pmatrix} 2 & 1 \\ t+1 & 2 \end{pmatrix} - \begin{pmatrix} 0 & 2 \\ 1 & -1 \end{pmatrix}^T$$
   is upper triangular but not diagonal.